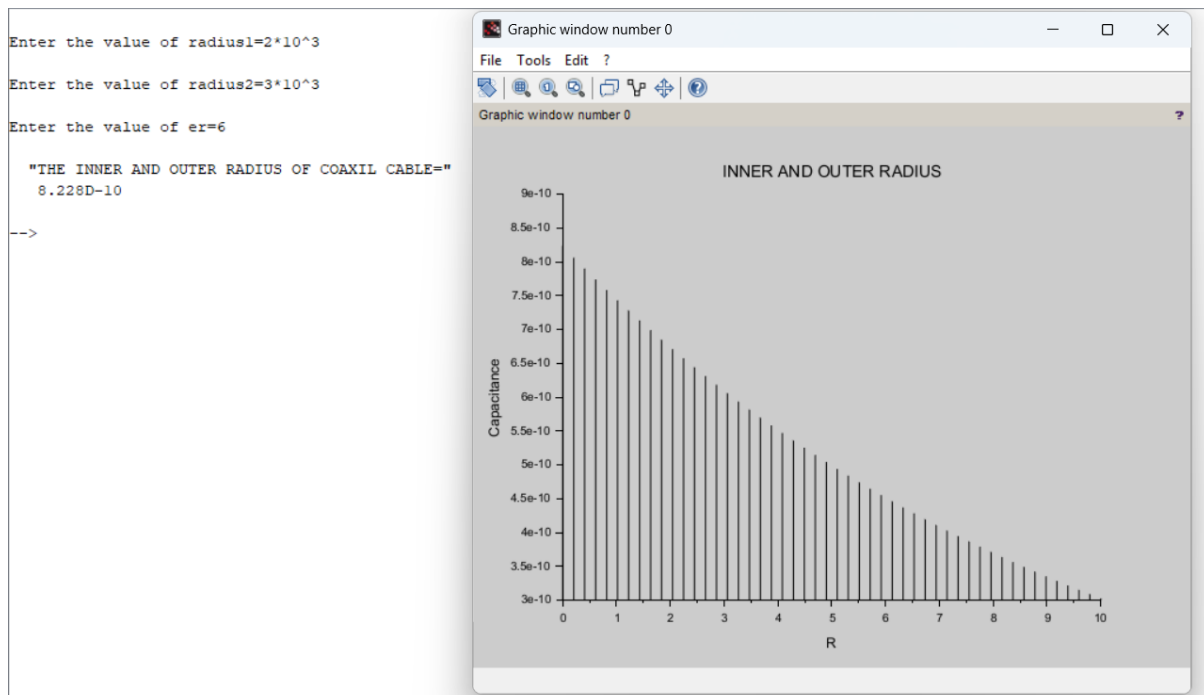


Exp no:8 -Determination and plotting of Capacitance of Co-axial cable per unit length with different (i) inner radius, (ii) outer radius and (iii) relative permittivity of the dielectric medium.

Theory:

```
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci)...
File Edit Format Options Window Execute ?
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
*exp333.sci X
1 clc;
2 clear;
3 r1=input("Enter the value of radius1=");
4 r2=input("Enter the value of radius2=");
5 er=input("Enter the value of er=");
6
7 r=r2/r1;
8 e0=8.854*10^-12;
9 pi=3.14;
10 e=e0*er;
11
12 c=(2*pi*e)/(log(r));
13 disp('THE INNER AND OUTER RADIUS OF COAXIL CABLE=',[c]);
14
15 x=linspace(0,10,50);
16 y=c*exp(-0.1*x);
17
18 figure();
19 subplot(1,1,1);
20 plot2d3(x,y);
21 xlabel("R");
22 ylabel("Capacitance");
23 title("INNER AND OUTER RADIUS");
```

Output:



Calculation:

Exp: 8

$r_1 = 2 \text{ mm}$
 $r_2 = 3 \text{ mm}$
 $l_r = 6$

The inner and outer radius of co-axial cable

$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$

$$C = \frac{2\pi\epsilon_0\epsilon_r}{\ln\left[\frac{r_2}{r_1}\right]}$$

$$C = \frac{2\pi \cdot 8.854 \times 10^{-12} \cdot 6}{\ln\left[\frac{3000}{2000}\right]}$$

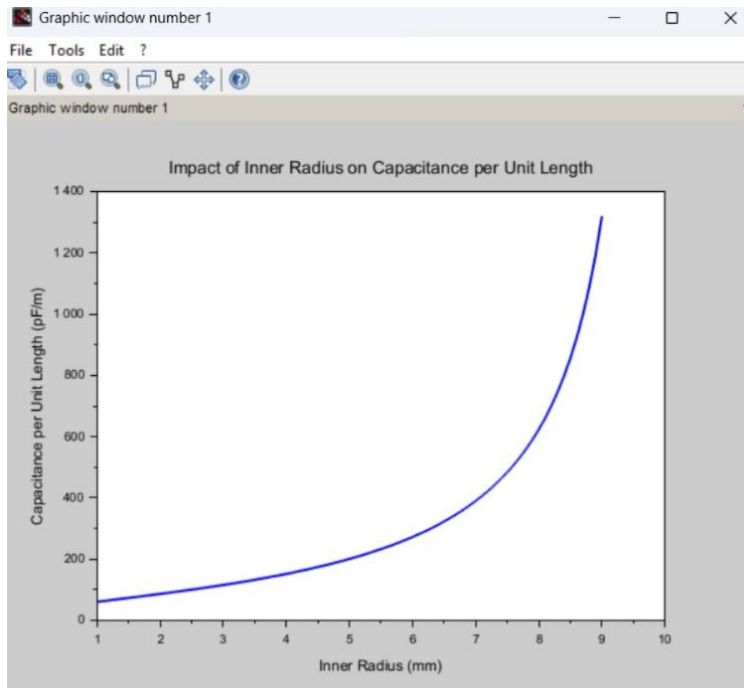
$$C = \frac{3.338 \times 10^{-10}}{\ln(1.5)}$$

$$C = 8.228 \times 10^{-10} \text{ F}$$

O/W

CASE STUDY 1: Variation with Inner Radius

```
exp 5.sci x exp 5 case 1.sci x exp 5 case ii.sci x exp 5 case iii.sci x exp 3.sci x exp
1 clc;
2 clear;
3 r1=input('Enter the value of radius1=');
4 r2=input('Enter the value of radius2=');
5 er=input('Enter the value of er=');
6
7 r=r2/r1;
8 e0=8.854*10^-12;
9 pi=3.14;
10 e=e0*er;
11
12 c=(2*pi*e)/(log(r));
13 disp('THE INNER AND OUTER RADIUS OF COAXIAL CABLE=', [c]);
14
15 x=linspace(0, 10, 50);
16 y=c*exp(-0.1*x);
17
18 figure();
19 subplot(1, 1, 1);
20 plot2d3(x,y);
21 xlabel('R');
22 ylabel('Capacitance');
23 title('INNER AND OUTER RADIUS');
24
25
```

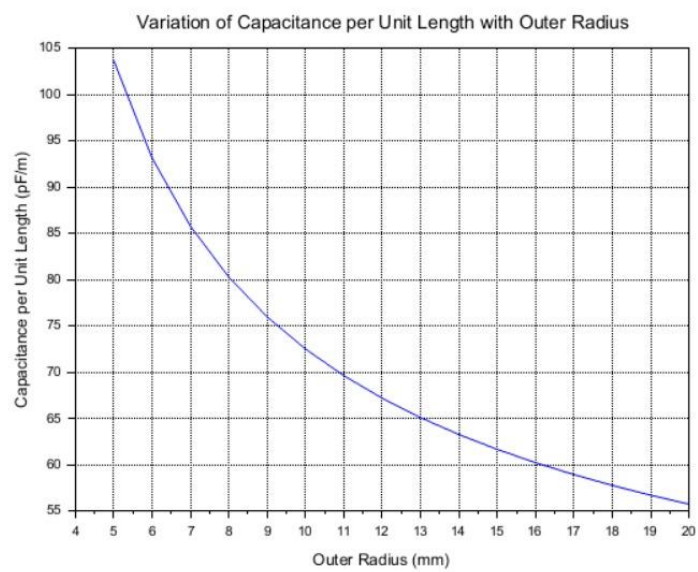
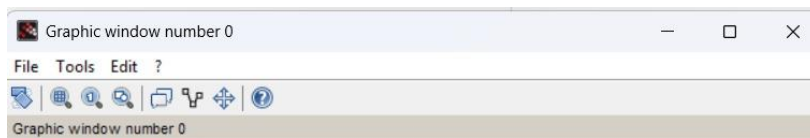


CASE STUDY 2: Variation with Outer Radius

```

1 //Clear the console and workspace
2 clc;
3 clear;
4 //Constants
5 epsilon0 = 8.854e-12; //Permittivity of free space (F/m)
6 epsilon_r = 3.0; //Relative permittivity
7 a = 1e-3; //Inner radius in meters (1 mm)
8 //Outer radius values (b) from 5 mm to 20 mm in steps of 1 mm
9 b_values = 5e-3:1e-3:20e-3;
10 //Initialize an array to store capacitance per unit length values
11 C_prime = zeros(size(b_values));
12 //Loop over the outer radius values and calculate capacitance per unit length
13 for i = 1:length(b_values)
14     b = b_values(i);
15     epsilon = epsilon0 * epsilon_r; //Permittivity of the dielectric
16     C_prime(i) = (2 * pi * epsilon) / log(b / a); //Capacitance per unit length
17 end
18 //Plotting the results
19 Scf(0); //Create a new figure
20 plot(b_values * 1e3, C_prime * 1e12); //Plot in mm and pF/m
21 xlabel('Outer Radius (mm)'); //Label for x-axis
22 ylabel('Capacitance per unit length (pF/m)'); //Label for y-axis
23 title('Variation of Capacitance per Unit Length with Outer Radius'); //Title of the plot
24 grid on; //Turn on the grid

```



CASE STUDY 3: Variation with Relative Permittivity

```
*exp 7 sci  X exp6 case i. sci  X exp6 case ii. sci  X exp6 case iii. sci  X exp7 c1. sci  X exp c2. sci  X exp8
1
2 clc;
3 clear;
4
5 %Constants
6 epsilon0 = 8.854e-12; %Permittivity of free space (F/m)
7 epsilon_r = 3.0; %Relative permittivity
8 a = 1e-3; %Inner radius (1 mm)
9
10 %Outer radius values (5 mm to 20 mm)
11 b_values = 5e-3:1e-3:20e-3;
12
13 %Initialize capacitance array
14 C_prime = zeros(b_values);
15
16 %Calculate capacitance per unit length
17 for i = 1:length(b_values)
18     b = b_values(i);
19     epsilon = epsilon0 * epsilon_r;
20     C_prime(i) = (2 * pi * epsilon) / log(b / a);
21 end
22
23 %Plot
24 scf(0);
25 plot(b_values * 1e3, C_prime * 1e12);
26
27 xtitle('Variation of Capacitance per Unit Length with Outer Radius', ...
28        'Outer Radius (mm)', ...
29        'Capacitance per Unit Length (pF/m)');
30
31 xgrid();
32
```

